

Poisson distribution

A discrete random variable X is said to follow Poisson distribution with parameter λ if its probability mass

function is
$$P(x) = \frac{e^{-\lambda} \lambda^x}{x!}, \quad x = 0, 1, 2, \dots, \infty$$

Poisson distribution as limiting form of Binomial distribution under the following conditions

- i) n , the number of trials is indefinitely large, i.e. $n \rightarrow \infty$.
- ii) p , the constant probability of success in each trial is very small. i.e., $p \rightarrow 0$.
- iii) $np (= \lambda)$ is finite (or) $p = \lambda / n$ and $q = 1 - \lambda / n$, where λ is a positive real number.

The following are some of the examples, which may be analysed using Poisson distribution.

1. The number of alpha particles emitted by a radioactive source in a given time interval.
2. The number of telephone calls received at a telephone exchange in a given time interval.

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3. The number of defective articles in a packet of 100.
4. The number of printing errors at each page of a book.
5. The number of road accidents reported in a city per day.

Moment generating function:

The MGF of the Poisson variate X with parameter λ is given as

$$\begin{aligned}
 M_x(t) &= E[e^{tx}] = \sum_{x=0}^{\infty} e^{tx} p(x) \\
 &= \sum_{x=0}^{\infty} e^{tx} \frac{e^{-\lambda} \lambda^x}{x!} \\
 &= e^{-\lambda} \sum_{x=0}^{\infty} \frac{(\lambda e^t)^x}{x!} \\
 &= e^{-\lambda} e^{\lambda e^t} \quad \because \sum_{x=0}^{\infty} \frac{a^x}{x!} = e^a \\
 &= e^{\lambda(e^t - 1)}
 \end{aligned}$$

Mean and variance of Poisson distribution

$$\begin{aligned}\mu_1^1 &= \left[\frac{d}{dt} M_x(t) \right]_{t=0} \\&= \left[\frac{d}{dt} e^{-\lambda} e^{\lambda e^t} \right]_{t=0} \\&= \left[e^{-\lambda} e^{\lambda e^t} \lambda e^t \right]_{t=0} \\&= e^{-\lambda} e^{\lambda} \lambda \\&= \lambda\end{aligned}$$

Mean of Poisson distribution = λ

$$\begin{aligned}\mu_2^1 &= \left[\frac{d^2}{dt^2} M_x(t) \right]_{t=0} \\&= \left[\frac{d}{dt} \left(\frac{d}{dt} e^{-\lambda} e^{\lambda e^t} \right) \right]_{t=0} \\&= \left[\frac{d}{dt} (e^{-\lambda} e^{\lambda e^t} \lambda e^t) \right]_{t=0} \\&= \lambda e^{-\lambda} [e^{\lambda e^t} \cdot e^t + e^t e^{\lambda e^t} \lambda e^t]_{t=0} \\&= \lambda e^{-\lambda} [e^{\lambda} + e^{\lambda} \lambda] \\&= \lambda e^{-\lambda} e^{\lambda} + \lambda^2 e^{-\lambda} e^{\lambda} \\ \mu_2^1 &= \lambda^2 + \lambda\end{aligned}$$

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Variance of Poisson distribution

$$\begin{aligned}\mu_2 &= \mu_2^1 - (\mu_1^1)^2 \\ &= \lambda^2 + \lambda - \lambda^2 = \lambda\end{aligned}$$

In Poisson distribution Mean = Variance = λ

Problems:

1. The number of monthly breakdowns of a computer is a random variable having a Poisson distribution with mean equal to 1.8. Find the probability that this computer will function for a month.

- a) Without a breakdown
- b) With only one breakdown and
- c) With atleast one breakdown

Let X denotes the number of breakdowns of the computer in a month.

X follows a Poisson distribution with mean $\lambda = 1.8$

$$\begin{aligned}P(X = x) &= \frac{e^{-\lambda} \lambda^x}{x!} \\ &= \frac{e^{-1.8} (1.8)^x}{x!}\end{aligned}$$

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$$a) p(x = 0) = e^{-1.8} = 0.1653$$

$$b) p(x = 1) = e^{-1.8}(1.8) = 0.2975$$

$$c) p(x \geq 1) = 1 - p(x = 0) = 0.8347$$

2. If the probability of a defective fuse from a manufacturing unit is 2%, in a box of 200 fuses, find the probability that a) exactly 4 fuses are defective b) more than 3 fuses are defective.

$$P=0.02, \quad n=200$$

$$\text{Mean} = \lambda = np = 200 * 0.02 = 4$$

$$\begin{aligned} a) P(X = 4) &= \frac{e^{-\lambda} \lambda^x}{x!} \\ &= \frac{e^{-4} (4)^4}{4!} = 0.1952 \end{aligned}$$

$$\begin{aligned} b) P(x > 3) &= 1 - P(x \leq 3) \\ &= 1 - [p(x = 3) + p(x = 2) + p(x = 1) + p(x = 0)] \\ &= 1 - e^{-4} \left[\frac{4^3}{3!} + \frac{4^2}{2!} + \frac{4^1}{1!} + \frac{4^0}{0!} \right] = 0.5669 \end{aligned}$$

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Fitting of Binomial Distribution

Fitting a binomial distribution means assuming that the given distribution is approximately binomial and hence finding the probability mass function and then finding the theoretical frequencies.

Problem:

Fit a binomial distribution for the following data:

x	0	1	2	3	4	5	6	Total
f	5	18	28	12	7	6	4	80

x	f	fx	Theoretical frequency
0	5	0	4
1	18	18	15
2	28	56	25
3	12	36	22
4	7	28	11
5	6	30	3
6	4	24	0
	$\sum f = N = 80$	$\sum fx = 192$	

$$\bar{x} = \frac{\sum fx}{\sum f} = \frac{192}{80} = 2.4$$

$$np = 2.4 \text{ (or) } 6p = 2.4$$

$$p = 0.4, \quad q = 0.6$$

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Theoretical distribution is given by

$$\begin{aligned}
 &= N \times P(X = x) \\
 &= 80 \times \sum_{x=0}^6 6_{c_x} p^x q^{n-x} \\
 &= 80 \times q^6 + 6_{c_1} q^{6-1} p + 6_{c_2} q^{6-2} p^2 + \dots \\
 &= 80 \times [(0.6)^6 + 6_{c_1} (0.6)^{6-1} (0.4) + 6_{c_2} (0.6)^{6-2} (0.4)^2 + 6_{c_3} (0.6)^{6-3} (0.4)^3 \\
 &\quad + 6_{c_4} (0.6)^{6-4} (0.4)^4 + 6_{c_5} (0.6)^{6-5} (0.4)^5 + 6_{c_6} (0.4)^6]
 \end{aligned}$$

$x:$	0	1	2	3	4	5	6
$f:$	3.73	14.93	24.88	22.12	11.06	2.95	0.33

Fitting of Poisson Distribution

1. Fit a Poisson distribution to the following data and calculate the theoretical frequencies.

Deaths	0	1	2	3	4
Frequency	122	60	15	2	1

x	f	fx	Theoretical frequency
0	122	0	121
1	60	60	61
2	15	30	15
3	2	6	3
4	1	4	0
	$N=200$	$\sum fx = 100$	

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$$\bar{x} = \frac{\sum fx}{\sum f} = \frac{100}{200} = 0.5$$

Theoretical distribution is given by

$$\begin{aligned} &= N \times P(X = x) \\ &= 200 \times \frac{e^{-\lambda} \lambda^x}{x!} \\ &= 200 \times \frac{e^{-0.5} (0.5)^x}{x!} \quad \text{putting } x = 0, 1, 2, 3, 4 \end{aligned}$$

$x:$	0	1	2	3	4
$f:$	121	61	15	3	0

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